

Ryan Zhenqi Zhou

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RESEARCH EXPERTISE

- GeoAI, Human Mobility, Disaster Resilience, Public Health, Spatial Data Science

AWARDS & HONORS

- 2025** CaGIS Student Scholarship (1-3 awardees nationwide per year), by *Cartography and Geographic Information Society (CaGIS)* [[Media Report](#)]
- 2025** Hugh W. Calkins Applied GIS Award, by *Department of Geography at the University at Buffalo – SUNY*
- 2025** AAG SAM Student Travel Award, by *AAG Spatial Analysis & Modeling Specialty Group (SAM)*
- 2024** CESASC Scholarship, by *Chinese-American Engineers and Scientists Association of Southern California (CESASC)*
- 2024** Michael Trapasso Climate Impacts Award, by *Department of Geography at the University at Buffalo - SUNY*
- 2024** Hugh W. Calkins Applied GIS Award, by *Department of Geography at the University at Buffalo - SUNY*
- 2023** Student Fellow, by *2023 NSF I-GUIDE Summer School Program*
- 2023** Hugh W. Calkins Applied GIS Award, by *Department of Geography at the University at Buffalo - SUNY*
- 2021** Outstanding Graduate Award, by *Nanjing Forestry University*
- 2020** National Scholarship Award, by *Department of Education, China*
- 2020** Merit Student Award, by *Nanjing Forestry University*
- 2020** First-class Scholarship Award, by *Nanjing Forestry University*
- 2019** Best Project Award, International Digital Urban Design Week held by *International Society of City & Regional Planners (ISOCARP)* and *Southeast University* in Nanjing

EDUCATION

- 2021 – Present** Ph.D. in GIScience
University at Buffalo - SUNY, USA
- Supervisor:** Dr. Yingjie Hu
- 2018 - 2021** Master of Engineering in Landscape Architecture
Nanjing Forestry University, China
- 2014 - 2018** Bachelor of Agriculture in Landscape Gardening
Zhejiang A&F University, China

EXTERNAL GRANTS

- 2022** **Student PI**, “Understanding community resilience in the 2021 Texas winter storm through human mobility data and its association with locations reported on social media”, Research Experience for Graduates (REG) Supplement Grant, **National Science Foundation**, 2022-2023, \$6000.
- 2020** **Student PI**, “Detecting the pedestrian shed and walking route environment of urban parks with open-source data: a case study in Nanjing, China”, Graduate Student Research Grant, *Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX20_0893)*, **Department of Education, Jiangsu, China**, 2020-2021, \$2300.

PUBLICATIONS

I have authored 23 peer-reviewed publications, including 5 first-author papers, 17 co-author papers, and 1 co-author refereed book chapter. According to Google Scholar, my publications have received 461 citations as of February 19, 2026.

Publications as the First Author

- 2025** **Zhou R.Z.**, Hu Y., Sun K., Muldoon R., Clark S., & Joseph K. (2025): Explainable GeoAI and statistical analysis reveal complementary insights about disparities of 311 help requests during the 2022 Buffalo blizzard. *International Journal of Disaster Risk Reduction*, 105635. <https://doi.org/10.1016/j.ijdr.2025.105635> [Impact Factor: 4.5]
- 2024** **Zhou R.Z.**, Hu Y., Zou L., Cai H., & Zhou B. (2024): Understanding the disparate impacts of the 2021 Texas winter storm and power outages through mobile phone location data and nighttime light images. *International Journal of Disaster Risk Reduction*, 103, 104339. <https://doi.org/10.1016/j.ijdr.2024.104339> [Impact Factor: 4.5]
- 2022** **Zhou R.Z.**, Hu Y., Tirabassi J.N., Ma Y., & Xu Z. (2022): Deriving neighborhood-level diet and physical activity measurements from anonymized mobile phone location data for enhancing obesity estimation. *International Journal of Health Geographics*, 21, 22. <https://doi.org/10.1186/s12942-022-00321-4> [Impact Factor: 3.2]
- 2021** **Zhou R.Z.**, Xu Z., Liu A., Zhou S., Mu L., & Zhang X. (2021): Mapping the accessibility of medical facilities of Wuhan during the COVID-19 pandemic. *International Journal of Geo-Information*, 10(5), 318. <https://doi.org/10.3390/ijgi10050318> [Impact Factor: 2.8]
- 2020** **Zhou R.Z.** & Xu Z. (2020): Detecting the pedestrian shed and walking route environment of urban parks with open-source data: a case study in Nanjing, China. *International Journal of Environmental Research and Public Health*, 17(13), 4826. <https://doi.org/10.3390/ijerph17134826> [Impact Factor: NA]

Publications as a Co-author

- 2025** Kim, J., Hu, Y., Elhami-Khorasani, N., Sun, K., & **Zhou, R. Z.** (2025): Assessment of deep learning models integrated with weather and environmental variables for wildfire spread prediction and a case study of the 2023 Maui fires. *Natural Hazards*, 122(1), 10. <https://doi.org/10.1007/s11069-025-07807-x> [Impact Factor: 3.7]
- 2025** Chen, Y., Nelson, J. K., Zhou, B., **Zhou, R. Z.**, Ye, S., Liu, H., Gu, Z., Kar, A., Kwon, H., Chen, P., Sun, M., & Kang, Y. (2025): Where are GIScience faculty hired from? Analyzing faculty mobility and research themes through hiring networks. *Cartography and Geographic Information Science*, 1-25. <https://doi.org/10.1080/15230406.2025.2567564> [Impact Factor: 2.4]
- 2025** Sun K., Hu Y., Joseph K., & **Zhou R.Z.** (2025): GALLOC: a GeoAnnotator for Labeling LOCation descriptions from disaster-related text messages. *International Journal of Geographical Information Science*, 1-31. <https://doi.org/10.1080/13658816.2025.2464643> [Impact Factor: 5.1]
- 2024** Sun K., **Zhou R.Z.**, Kim J., & Hu Y. (2024): PyGRF: An improved Python Geographical Random Forest model and case studies in public health and natural disasters. *Transactions in GIS*, 28(7), 2476-2491. <https://doi.org/10.1111/tgis.13248> [Impact Factor: 2.3]
- 2024** Tirabassi, J. N., Wang, J., **Zhou, R. Z.**, & Hu, Y. (2024): Human mobility data demonstrates increase in park visitation since start of COVID-19 pandemic in Buffalo, New York. *Preventive Medicine Reports*, 102650. <https://doi.org/10.1016/j.pmedr.2024.102650> [Impact Factor: 2.4]
- 2023** Sun K., Hu Y., Lakhanpal, G., & **Zhou, R.Z.** (2023): Spatial cross-validation for GeoAI. *Handbook of Geospatial Artificial Intelligence*, Taylor & Francis Group.
- 2023** Hu, Y., Mai, G., Cundy, C., Choi, K., Lao, N., Liu, W., Lakhanpal, G., **Zhou, R.Z.**, & Joseph, K. (2023): Geo-knowledge-guided GPT models improve the extraction of location descriptions from disaster-related social media messages. *International Journal of Geographical Information Science*, 1-30. <https://doi.org/10.1080/13658816.2023.2266495> [Impact Factor: 5.1]
- 2023** Sun K., Hu Y., Ma Y., **Zhou R. Z.**, & Zhu Y. (2023): Conflating point of interest (POI) data: a systematic review of matching methods. *Computers, Environment and Urban Systems*, 103, 101977. <https://doi.org/10.1016/j.compenvurbsys.2023.101977> [Impact Factor: 8.3]
- 2023** Xu Z., Li M., Chen Y., & **Zhou R.Z.** (2023): Assessing the cycling accessibility of comprehensive parks with online maps: a case study of Nanjing City. *Journal of Nanjing Forestry University Natural Science*, 1-8 (in Chinese).
- 2022** Zhang B., Dong Y., Kelobonye K., **Zhou R.Z.**, & Xu Z. (2022): Delineating walking catchment of the existing and proposed public sports facilities with open-source data: a case study of Nanjing. *Applied Spatial Analysis and Policy*, 16, 729-749. <https://doi.org/10.1007/s12061-022-09499-3> [Impact Factor: 1.9]
- 2022** Chen Y., Zhang B., Li M., **Zhou R.Z.**, & Xu Z. (2022): Concatenating daily exercise routes with public sports facilities, bicycle lanes, and green spaces: a feasibility analysis in Nanjing, China. *Land*, 11, 2251. <https://doi.org/10.3390/land11122251> [Impact Factor: 3.2]

- 2022 Yue M., Hu Y., Moncrieff G.R., Slingsby J.A., Wilson A.M., Maitner B., & **Zhou R.Z.** (2022): Forecasting vegetation dynamics in an open ecosystem by integrating deep learning and environmental variables. *International Journal of Applied Earth Observation and Geoinformation*, 114, 103060. <https://doi.org/10.1016/j.jag.2022.103060> [Impact Factor: 8.6]
- 2022 Dong Y., Zhang B., **Zhou R.Z.**, & Xu Z. (2022): Assessing the accessibility of swimming pools in Nanjing by walking and cycling using Baidu Maps. *International Journal of Geo-Information*, 11(10), 515. <https://doi.org/10.3390/ijgi11100515> [Impact Factor: 2.8]
- 2021 Xu Z., **Zhou R.Z.**, Wang Y., Han L., & Xing J. (2021): Analysis on walking shed in comprehensive parks and urban form from a Park-City perspective. *City Planning Review*, 45(03), 81-90 (in Chinese).
- 2021 Xu Z., Liu A., **Zhou R.Z.**, & Han L. (2021): Analyzing and scenarioizing walking routes and urban context of middle school students with Baidu Map: a case study of Jianye District of Nanjing. *Modern Urban Research*, 2, 33-40 (in Chinese).
- 2021 Xu Z., Gao Z., **Zhou R.Z.**, & Zhang J. (2021): Research on campus sitting space: a case study of Nanjing Forestry University. *Modern Urban Research*, 4, 30-35 (in Chinese).
- 2020 Liu A., Kelobonye K., **Zhou R.Z.**, Xu Q., Xu Z., & Han L. (2020): School commuting mode shift: a scenario analysis for active school commuting using GIS and online map API. *International Journal of Geo-Information*, 9(9), 520. <https://doi.org/10.3390/ijgi9090520> [Impact Factor: 2.8]
- 2020 Zhang J., **Zhou R.Z.**, & Xu Z. (2020): Suitability evaluation of university facilities layout based on demand characteristics: taking several universities in Nanjing as examples. *Modern Urban Research*, 4, 43-51 (in Chinese).

Manuscripts in Preparation

- 2026 **Zhou R.Z.**, Hu Y., Zhang L., Muldoon R., Clark S., & Joseph K. (2026): Assessing spatial disparities of human mobility disruptions during the 2022 Buffalo blizzard through mobile phone location data and transformer model, *In preparation*.

CONFERENCE & WORKSHOP PRESENTATIONS

- 2025 **Oral presentation:** Explainable GeoAI and statistical analysis reveal complementary insights about disparities of 311 help requests during the 2022 Buffalo blizzard, in the *2025 AAG Annual Meeting*, Mar. 26, 2025, Detroit, USA.
- 2024 **Poster presentation:** Understanding the disparate impacts of the 2021 Texas winter storm and power outages through mobile phone location data and nighttime light images, in the *31st International Conference on Geoinformatics, CPGIS*, Aug. 15, 2024, Toronto, Ontario, Canada.
- 2023 **Oral presentation:** Improving the 3D representation of rivers in digital elevation models (DEM), in the *2023 NSF I-GUIDE Summer School*, Aug. 11, 2023, Boulder, Colorado, USA.
- 2023 **Oral presentation:** Understanding spatial and temporal impacts of the Texas winter storm in 2021 via mobile phone location data and nighttime light images, in the *2023 AAG Annual Meeting*, Mar. 26, 2023, Denver, Colorado, USA.
- 2022 **Oral presentation:** Deriving neighborhood-level diet and physical activity measurements from anonymized mobile phone location data for enhancing obesity estimation, in the *GEOMED 2022*, Oct. 14, 2022, University of California, Irvine, California, USA.
- 2022 **Poster presentation:** Human mobility data demonstrates increase in park visitation since start of COVID-19 pandemic in Buffalo, New York, in the *American College of Preventive Medicine (ACPM) 2022 annual meeting*, Jun. 15-18, 2022, Denver, Colorado, USA (Co-author).
- 2022 **Poster presentation:** Mapping the accessibility of medical facilities of Wuhan the COVID-19 pandemic, in the *University Consortium for Geographic Information Science (UCGIS) Symposium 2022*, Jun. 7-9, 2022, Syracuse, New York, USA.
- 2022 **Oral presentation:** The role of place visits related to diet and physical activity from mobile phone location data in enhancing obesity estimation at the neighborhood level, in the *Graduate Student Lightning Talks, Spring 2022 Colloquium Speaker Series, UB Department of Geography*, Feb. 18, 2022, Buffalo, New York, USA.
- 2019 **Poster presentation:** Measuring the walkability around the urban parks: a case study of Nanjing, in the *16th International Conference on Urban Health*, Nov. 4-8, 2019, Xiamen, Fujian, China.
- 2019 **Poster presentation:** Mapping the service area and population of urban parks for park city initiative: a case study of Nanjing, in the *16th International Conference on Urban Health*, Nov. 4-8, 2019, Xiamen, Fujian, China.

- 2019** **Oral presentation:** Old community renovation based on data analysis, in the *9th Summer Workshop DigitalFUTURES YOUNG Forum 2019—Architectural Intelligence held at the School of Architecture and Urban Planning (CAUP), Tongji University*, Jun. 30 to Jul. 6, 2019, Shanghai, China.
- 2019** **Poster presentation:** Intelligence and Strategy of Urban Deduction—Spark Planning (UrbanGAN), in the *9th Summer Workshop DigitalFUTURES SHANGHAI 2019—Architectural Intelligence held at the School of Architecture and Urban Planning (CAUP), Tongji University*, Jun. 30 - Jul. 6, 2019, Shanghai, China (Co-author).
- 2019** **Oral presentation:** Social Loop, loops to connect city and communities, in the *International Digital Urban Design Week held by International Society of City & Regional Planners (ISOCARP) and Southeast University*, Jun. 10-15, 2019, Nanjing, Jiangsu, China.
- 2018** **Poster presentation:** The urban community agriculture practices in Chinese cities: case studies in Nanjing, in the *15th International Conference on Urban Health*, Nov. 27-30, 2018, Kampala, Uganda, Africa (Co-author).

PROFESSIONAL EXPERIENCE

- August 2021 – Present** **GeoAI Lab**, University at Buffalo – SUNY, Research Assistant
I conduct research on geospatial data science and AI with applications in the areas of disaster resilience (e.g., winter storms) and public health (e.g., obesity estimation)
- May 2024 – Present** **World Bank – IFC**, Global Health, Geospatial Data Scientist | Consultant
Research Project: A geospatial analysis of SafeCare healthcare facilities characteristics, climate risks, and economic and demographic factors in Kenya and Tanzania
- August 2024 – June 2025** **Cornell University**, Department of Population Health Sciences, Research Assistant
Research Project: Using machine learning to identify social risk factors of hypertension and diabetes in New York City: evidence to support the HealthyNYC initiative
- August 2023 – August 2023** **NSF I-GUIDE Summer School**, Student Fellow
Research Project: Improving the 3D representation of rivers in digital elevation models (DEM)

TEACHING EXPERIENCE

- August 2022 – May 2023** Department of Geography, University at Buffalo – SUNY, Teaching Assistant
GEO 481/506 Geographical Information Systems Lab
Student Evaluation: 4.75/5 (Gold Standard)

TECHNICAL SKILLS

- **Programming/Tools:** Python, ArcGIS Pro, R, SQL, Git
- **Python Libraries:** NumPy, Pandas, GeoPandas, Shapely, Fiona, Rasterio, Folium, SciPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, Seaborn, Bokeh, Request, Json, PyQt
- **Models:** Regression (OLS, GWR, MGWR, etc.), Tree-based Models (RF, XGBoost, GRF), Neural Networks (DNN, etc.), Time Series Models (LSTM, Transformer, etc.), Explainable AI (SHAP), Generative AI (GPT), Clustering (K-means)

SERVICE

Peer Review for Academic Journals

- Computational Urban Science
- Environment and Planning B: Urban Analytics and City Science
- Journal of Map
- Royal Society Open Science
- International Journal of Digital Earth
- Transactions in GIS
- Scientific Data

- Spatial and Spatio-temporal Epidemiology
- International Journal of Disaster Risk Reduction
- Urban Informatics
- iScience
- PLOS One
- International Journal of Geographical Information Science
- Big Earth Data

Voluntary Service

- Co-host of GeoAI for Disaster Resilience session at AAG, Mar. 26, 2025, USA
- Host of GISalon, GISphere (International Student Academic Organization), Oct. 2021-Jul. 2024, USA
- Faculty Liaison for PhD, Geography GSA, University at Buffalo - SUNY, Sept. 2022-May. 2024, USA
- Conference Assistant, 2024 CESASC 62nd Annual Convention, May 4, 2024, USA
- Conference Assistant, 2023 AAG Annual Meeting, Mar. 23-27, 2023, USA

Media Coverage

UB Now: [UB PhD student receives national cartography and geographic information society scholarship](#)

Channel 2 WGRZ: [Study looks at social media's role in connecting the community during the 2022 blizzard](#)

Buffalo News: [Facebook groups saved lives during '22 blizzard. UB studying how they kept communities connected](#)